K-theory and the Brouwer Fixed Point Theorem

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1 Introduction

This expository paper serves to introduce the fundamentals of topological K-theory, aiming to provide a K-theoretic proof of the Brouwer fixed point theorem. The exposition unfolds in three distinct parts.

In the first section, we present the construction of vector bundles through a continuous functor as in [1, p. 6–9], defining operations such as $E \oplus F$, $E \otimes F$, and $\mathrm{Hom}(E, F)$ within this framework. Additionally, we introduce pullback bundles, proving that homotopic maps from a paracompact space yield isomorphic pullback bundles.

In the second section, we construct K-theory, reduce K-theory for arbitrary topological spaces, and verify their functorial properties. Based on the theorem presented in the first section, we also prove that homotopically equivalent paracompact spaces inherently possess isomorphic K-theory.

In the third part, we state the Bott periodicity theorem and cite [2, p. 41–51] as the reference for the proof. Then we compute the reduced K-theory of S^n. Finally, with all these tools, we present a proof of the Brouwer fixed point theorem using K-theory.

We have two main references: one is the book "Vector Bundles and K-Theory" [2] by Allen Hatcher and the other one is "K-theory" [1] by Michael Atiyah.

2 Basic Definition of Vector Bundles

In this section, we want to introduce and define some definitions/properties that we will use later in this paper. We first introduce the definition of vector bundles.

Definition 2.1. A complex vector bundle is a map $p : E \to X$ together with a complex vector space structure on $p^{-1}(x)$ for each $x \in X$, such that the following local triviality condition is satisfied: There is a cover of X by open sets U_α for each of which there exists a homeomorphism $h_\alpha : p^{-1}(U_\alpha) \to U_\alpha \times \mathbb{C}^n$ taking $p^{-1}(x)$ to $\{x\} \times \mathbb{C}^n$ by a complex vector space isomorphism for each $x \in U_\alpha$. Such an h_α is called a local trivialization of the vector bundle. The space X is called the base space, E is the total space, and the vector spaces $p^{-1}(x)$ are the fibers.

Like other mathematical objects, we also have homomorphism/isomorphism between vector bun-

dles.

Definition 2.2. A homomorphism from one family $p : E \to X$ to another family $q : F \to X$ is a continuous map $f : E \to F$ such that:

(i) $qf = p$

(ii) For each $x \in X$, $f : E_x \to F_x$ is a complex linear map

Definition 2.3. We say that f is an isomorphism if f is bijective and f^{-1} is continuous. If there exists an isomorphism between E and F, we say that they are isomorphic.

To check if a map is a vector bundle isomorphism, we use the following lemma:

Lemma 2.1. *A continuous map $h : E_1 \to E_2$ between vector bundles over the same base space X is an isomorphism if it takes each fiber $p_1^{-1}(x)$ to the corresponding fiber $p_2^{-1}(x)$ by a complex linear isomorphism.*

The proof is presented in [2][p. 8, lemma 1.1].

The abundance of operations on vector spaces, e.g., direct sum, tensor product, dual, etc., motivates us to generalize them to vector bundles. For example, if we have E as a vector bundle, to define E^* the idea is to replace each fiber E_x by its dual E_x^*, then define the topology carefully.

Formally speaking, denote **FinCVect** the category of finite complex dimensional vector spaces. Let $\mathbf{FinCVect}^n$ be the n-fold product category of **FinCVect**. Fix a functor $T : \mathbf{FinCVect}^n \to \mathbf{FinCVect}$. In the following discussion, one assumes the topology on finite dimensional vector spaces to be the standard one which makes any isomorphism to $\mathbb{C}^n$ a homeomorphism.

Definition 2.4. T is *continuous* if for any $V_1, ..., V_n, W_1, ..., W_n \in \mathbf{FinCVect}$, the map $T : \mathrm{Hom}(V_1, W_1) \times ... \times \mathrm{Hom}(V_n, W_n) \to \mathrm{Hom}(T(V_1, ..., V_n), T(W_1, ..., W_n))$ is continuous.

Example 2.4.1. Consider the dual functor $* : \mathbf{FinCVect} \to \mathbf{FinCVect}$ sending V to V^* and for complex linear map between finite complex vector spaces $f : V \to W$, $f^* : W^* \to V^*$. $T : \mathrm{Hom}(V, W) \to \mathrm{Hom}(W^*, V^*)$ is a linear transformation, so is a continuous functor.

Definition 2.5. Let X be a topological space. Let $p_i : E_i \to X$ be n complex vector bundles, we define the set $T(E_1, ..., E_n)$ to be the union

$$\bigsqcup_{x\in X} T((E_1)_x, ..., (E_n)_x)$$

and, if $\psi_i : E_i \to F_i$, we define $T(\psi_1, ..., \psi_n) : T(E_1, ..., E_n) \to T(F_1, ..., F_n)$ by the maps $T(\psi_x) : T((E_1)_x, ..., (E_n)_x) \to T((F_1)_x, ..., (F_n)_x)$.

Let $(U, \varphi_i : p_i^{-1}(U) \to U \times \mathbb{C}^{m_i})$ be a local trivialization of E_i. For each $x \in X$, denote $(\varphi_i)_x : (E_i)_x \to \mathbb{C}^{m_i}$ the isomorphism restricted to the fiber of x and then define $\varphi_x = ((\varphi_1)_x, ..., (\varphi_n)_x)$. $T(\varphi_x) \in \mathrm{Hom}(T((E_1)_x, ..., (E_n)_x), T(\mathbb{C}^{m_1}, ..., \mathbb{C}^{m_n}))$. Since $T(\mathbb{C}^{m_1}, ..., \mathbb{C}^{m_n}))$ is a finite complex dimensional vector space, there exists an isomorphism $S : T(\mathbb{C}^{m_1}, ..., \mathbb{C}^{m_n})) \to \mathbb{C}^k$ for some k.

One has the following bijection Φ_φ:

$$\Phi_\varphi : \bigsqcup_{x\in U} T((E_1)_x, ..., (E_n)_x) \to \bigsqcup_{x\in U} T(\mathbb{C}^{m_1}, ..., \mathbb{C}^{m_n}) \cong U \times \mathbb{C}^k$$

$$(x, v) \mapsto (x, S \circ \varphi_x(v))$$

One would want to topologize $T(E_1, ..., E_n)$ so that these bijections become local trivializations. In fact, one has the following theorem.

Theorem 2.2. *$T(E_1, ..., E_n)$ has a unique topology so that $p : T(E_1, ..., E_n) \to X$ is a complex vector bundle with trivializations (U, Φ_φ) as in the above construction. Under this topology, if $\psi_i : E_i \to F_i$ are complex vector bundle homomorphisms, $T(\psi_1, ..., \psi_n) : T(E_1, ..., E_n) \to T(F_1, ..., F_n)$ is continuous.*

Sketch of Proof. We prove the result in the case when $T : \mathbf{FinCVect} \to \mathbf{FinCVect}$, the cases of more variables can be proved in a similar way. Let V & W be complex vector spaces.

We begin by defining $T(E)$ in the case that E is a product bundle. If $E = X \times V$, we identify $T(E)$ as $X \times T(V)$ with the product topology. Suppose that $F = X \times W$ and that $\psi : E \to F$ is a vector bundle homomorphism, then $\psi : X \to \mathrm{Hom}(V, W), x \mapsto \psi_x$ is continuous. Let $\varphi : X \to \mathrm{Hom}(V, W)$ be the corresponding map. By continuity, $T : \mathrm{Hom}(V, W) \to \mathrm{Hom}(T(V), T(W))$ is continuous, so $T\psi : X \to \mathrm{Hom}(T(V), T(W))$ is continuous. Thus $T(\psi) : X \times T(V) \to X \times T(W)$ is also continuous. If ψ is an isomorphism, then $T(\psi)$ will be an isomorphism since it is continuous and an isomorphism on each fiber.

Now suppose that E is trivial, but has no preferred product structure. Choose a vector bundle isomorphism $\alpha : E \to X \times V$ and topologize $T(E)$ by requiring $T(\alpha) : T(E) \to X \times T(V)$ to be a homeomorphism. If $\beta : E \to X \times W$ is any other isomorphism, by letting $\psi = \beta\alpha^{-1}$ above, we see that $T(\alpha)$ and $T(\beta)$ induce the same topology on $T(E)$, since $T(\psi) = T(\beta)T(\alpha)^{-1}$ is a homeomorphism. Thus, the topology on E does not depend on the choice of α. Further, if $Y \subset X$, it is clear that the topology on $T(E)|_Y$ is the same as that on $T(E|_Y)$. Finally, if $\psi : E \to F$ is a homomorphism of trivial bundles, we see that $T(\psi) : T(E) \to T(F)$ is continuous, and therefore is a vector bundle homomorphism.

Now suppose that E is any vector bundle. Then if $U \subset X$ is such that $E|_U$ is trivial, we topologize $T(E|_U)$ as above. We topologize $T(E)$ by taking for the open sets those subsets $V \subset T(E)$ such that $V \cap T(E)|_U$ is open in $T(E|_U)$ for all open $U \subset X$ for which $E|U$ is trivial. One can verify that if $Y \subset X$, the topology on $T(E|_Y)$ is the same as that on $T(E)|_Y$, and that, if $\psi : E \to F$ is any homomorphism, $T(\psi) : T(E) \to T(F)$ is also a homomorphism. $\square$

Lemma 2.3. *Given two vector bundles $p_E : E \to X$ and $p_F : F \to X$,*

$$E \oplus F \cong F \oplus E$$
$$E \otimes F \cong F \otimes E$$

Proof. Consider the function

$$g : E \oplus F \to F \oplus E$$
$$(x, (e, f)) \mapsto (x, (f, e))$$

This function is bijective since on each fiber we have an isomorphism. It suffices to prove continuity and apply 2.1.

For each point $x \in X$, there exists an open neighborhood restricted on which E and F are both trivial. Let $\varphi_E : p_E^{-1}(U) \to U \times \mathbb{C}^n$ be a trivialization of E and induces on each fiber a complex linear isomorphism $(\varphi_E)_x : p_E^{-1}(x) \to \mathbb{C}^n$ and $\varphi_F : p_F^{-1}(U) \to U \times \mathbb{C}^m$ be a trivialization of F and induces on each fiber a complex linear isomorphism $(\varphi_F)_x : p_F^{-1}(x) \to \mathbb{C}^m$.

According to 2.2, $E \oplus F$ and $F \oplus E$ have trivializations on U defined as

$$\varphi_{E\oplus F} : E \oplus F \longrightarrow U \times \mathbb{C}^n \times \mathbb{C}^m,\ (x,(e,f)) \mapsto (x,(\varphi_E)_x(e),(\varphi_F)_x(f))$$
$$\varphi_{F\oplus E} : F \oplus E \longrightarrow U \times \mathbb{C}^m \times \mathbb{C}^n,\ (x,(f,e)) \mapsto (x,(\varphi_F)_x(f),(\varphi_E)_x(e))$$

Under these trivializations, $\tilde{g} := \varphi_{F\oplus E} \circ g \circ \varphi_{E\oplus F} : U \times \mathbb{C}^n \times \mathbb{C}^m \to U \times \mathbb{C}^m \times \mathbb{C}^n$ satisfies:

$$\begin{aligned}
\tilde{g}(x,v,w) :&= \varphi_{F\oplus E} \circ g \circ \varphi_{E\oplus F}(x,v,w) \\
&= \varphi_{F\oplus E} \circ g(x,(\varphi_E)_x^{-1}(v),(\varphi_F)_x^{-1}(w)) \\
&= \varphi_{F\oplus E}(x,(\varphi_F)_x^{-1}(w),(\varphi_E)_x^{-1}(v)) \\
&= (x,w,v)
\end{aligned}$$

is a continuous function. Because $\varphi_{E\oplus F}$ and $\varphi_{F\oplus E}$ are homeomorphisms, g is continuous.

The case for tensor product is completely analogous. □

Then, we want to define the pullbacks of vector bundles and state the elementary properties of pullbacks.

Proposition. *[1, p.18, proposition 1.5] Given a map $f : Y \to X$ and a vector bundle $p : E \to X$, there exists a unique vector bundle (up to vector bundle isomorphism) $p' : E' \to Y$ with a map $f' : E' \to E$ taking the fiber of E' over each point $y \in Y$ isomorphically onto the fiber of E over $f(y)$.*

Definition 2.6. The bundle satisfying the above property is called the *pullback* of E by f.

Proposition (Functoriality)**.** *[2, p.20] $(fg)^*(E) \simeq g^*(f^*(E))$ and $\mathbb{1}^*(E) \simeq E$.*

Proposition. *A vector bundle homomorphism $\varphi : E \to F$ can be identified with a section of* $\mathrm{Hom}(E,F)$.

Fix a topological space X.

Definition 2.7. Let $\{U_\alpha\}_{\alpha\in A}$ be an open cover. A *refinement* of $\{U_\alpha\}_{\alpha\in A}$ is an open cover $\{V_\beta\}_{\beta\in B}$ so that for any $\beta \in B$, there exists some $\alpha \in A$ so that $V_\alpha \subset U_\beta$. An open cover $\{U_\alpha\}_{\alpha\in A}$ is *locally finite* if for any $x \in X$, there exists an open neighborhood W_x of x so that $W_x \cap U_\alpha$ for finitely many α. X is *paracompact* if each cover admits a locally finite refinement.

Definition 2.8 (Support)**.** Let $f : X \to K \subset \mathbb{C}$. $\mathrm{Supp}(f) = \overline{\{x : f(x) \neq 0\}} \subset X$ is the *support* of f. f is *compactly supported* if $\mathrm{Supp}(f)$ is compact.

Definition 2.9 (Partitions of Unity)**.** Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of X. Then a collection of maps $\{\varphi_\alpha : X \to [0,1]\}_{\alpha \in A}$ is said to be *a partition of unity subordinate to the open cover* $\{U_\alpha\}_{\alpha \in A}$ if

(i). for any $\alpha \in A$, $\mathrm{Supp}(\varphi_\alpha) \subset U_\alpha$;

(ii). each point $x \in X$, there exists a neighborhood V_x of x so that V_x intersects $\mathrm{Supp}\,\varphi_\alpha$ for finitely many α;

(iii).$\sum_{\alpha \in A} \varphi_\alpha = 1$. This sum has finitely many terms because of (ii).

Theorem 2.4. *[3, p.42-43, theorem 2.23] If X is paracompact Hausdorff and $\{U_\alpha\}_{\alpha \in A}$ is an open cover. Then there exists a partition of unity subordinate to $\{U_\alpha\}_{\alpha \in A}$.*

Lemma 2.5. *Let $Y \subset B$ be a closed subset of a paracompact Hausdorff space and $p : E \to B$ be a vector bundle. Let $s : Y \to E$ be a section, then there exists a section $t : U \to E$, $Y \subset U$ open, so that $t|_Y = s$.*

Sketch of Proof. Let $(U_\alpha, \varphi_\alpha)$ be trivializations of B and (ψ_n) be a partition of unity subordinate to (U_α). Then $s_n := \varphi_n \circ s : Y \cap U_n \to Y \times \mathbb{C}^m$ can be extended to the whole U_n by Tietze extension theorem. Take $t := \sum \psi_n s_n$. □

Theorem 2.6. *Let A be a paracompact Hausdorff space, B a topological space, and $p : E \to B$ be a vector bundle. If $f : A \times I \to B$ is a homotopy, then $f_0^* E$ is isomorphic to $f_1^* E$.*

Proof. Consider f^*E and $\pi^* f_t^* E$, then define

$$\varphi : f^*E|_{\{t\} \times Y} \to \pi^* f_t^* E|_{\{t\} \times Y}$$
$$((t,y), v) \mapsto ((t,y), v)$$

This is a vector bundle isomorphism, which is a section of $\mathrm{Hom}(f^*E, \pi^* f_t^* E)|_{\{t\} \times Y}$. By previous lemma, there is a section $\tilde{\varphi} : X \to \mathrm{Hom}(f^*E, \pi^* f_t^* E)$ extending φ. For each $x \in X$, take $(U, \varphi_\varphi^\psi)$ a

local trivialization of $\mathrm{Hom}(f^*E, \pi^* f_t^* E)$, then for $y \in U$, $\varphi(x)$ is an isomorphism if and only if

$$U \xrightarrow{\varphi} \mathrm{Hom}(f^*E, \pi^* f_t^* E)|_U \xrightarrow{\varphi^{\psi}_{\varphi}|_{p^{-1}(U)}} U \times \mathrm{Hom}(\mathbb{C}^n, \mathbb{C}^n) \xrightarrow{\pi_2} \mathrm{Hom}(\mathbb{C}^n, \mathbb{C}^n) \xrightarrow{\det} \mathbb{C}$$

is nonzero at x. Since the composition is continuous and x is sent to a nonzero value, there exists an open neighborhood V of x so that $\varphi(y)$ is an isomorphism for each $y \in V$. Hence, $\tilde{\varphi}$ is an isomorphism on an open neighborhood V of $\{t\} \times Y$, so must contain an open strip $\delta_t \times Y$ where δ_t is an open interval containing t.

Let

$$\mathcal{F} := \{t \in I : f_t^* E \cong f_0^* E\}$$

then $0 \in \mathcal{F}$. By previous argument, there exists an open interval δ_t containing t so that for each $t' \in \delta_t$, $f_{t'}^* E \cong f_t^* E \cong f_0^* E$, so $\delta_t \subset \mathcal{F}$, so $\mathcal{F}$ is open. If $t \notin \mathcal{F}$, by previous argument, there exists an open interval δ_t containing t so that for each $t' \in \delta_t$, $f_{t'}^* E \cong f_t^* E \ncong f_0^* E$, so $\delta_t \subset I - \mathcal{F}$, so $I - \mathcal{F}$ is open, i.e., $\mathcal{F}$ is closed. by connectedness of I, $\mathcal{F} = I$. □

3 Construction of the Functor K and $\widetilde{K}$

3.1 The Functor Vect : Top → CRig

Definition 3.1. A triple $(S, +, \times)$ is called a commutative semiring if $(S, +)$ is a commutative monoid with identity 0, $(S, \times)$ is a commutative monoid with identity 1, and $\times$ is distributive, and 0 annihilates everything. $f : S \to S'$ is called a semiring homomorphism if $f(a + b) = f(a) + f(b)$ and $f(ab) = f(a)f(b)$ and $f(0) = 0$ and $f(1) = 1$. The category of commutative semirings is denoted as **CRig**, and the category of commutative rings is denoted as **CRing**.

Let $X, Y \in$ **Top**.

Define $\mathrm{Vect}(X)$ as the isomorphism classes of complex vector bundles over X.

Proposition. *If $E \cong E'$ and $F \cong F'$, then $E \oplus F \cong E' \oplus F'$ and $E \otimes F \cong E' \otimes F'$.*

Proof. If $f : E \to E'$ and $g : F \to F'$ are two vector bundle isomorphisms and f^{-1}, g^{-1} are their inverses, then $T(f, g) : T(E, F) \to T(E', F')$ is a bundle homomorphism and has the inverse

$T(f^{-1}, g^{-1})$, so $T(E, F) \cong T(E', F')$. Just let T be $\oplus$ and $\otimes$. $\square$

Therefore, $\oplus$ and $\otimes$ are well-defined binary operations on $\mathrm{Vect}(X)$. In fact, these operations make $\mathrm{Vect}(X)$ into a commutative semiring, as we defined in 3.1.

Proposition. $(\mathrm{Vect}(X), \oplus, \otimes)$ *is a commutative semiring with the identities* $\mathbb{1}_{\oplus} = \epsilon^0$*, the trivial bundle of zeroth dimension, and* $\mathbb{1}_{\otimes} = \epsilon^1$*, the trivial bundle of first dimension.*

Proof. By 2.3, $E \oplus F \cong F \oplus E$, for any $[E], [F] \in \mathrm{Vect}(X)$ $[E] \oplus [F] = [E \oplus F] = [F \oplus E] = [F] \oplus [E]$ and $[E] \oplus [\epsilon^0] \cong [E]$, $(\mathrm{Vect}(X), \oplus)$ is a commutative monoid with identity ϵ^0. Similarly, $(\mathrm{Vect}(X), \otimes)$ is a commutative monoid with identity ϵ^1. Notice $\otimes$ is distributive and ϵ^0 annihilates everything. So $(\mathrm{Vect}(X), \oplus, \otimes)$ forms a commutative semirings. $\square$

Proposition. *If* $F \cong F'$*,* $f^*F \cong f^*F'$*.*

Proof. Let $g : F \to F'$ be a vector bundle isomorphism. Define for each $x \in X$ and $v \in F_{f(x)}$, $\tilde{g}(v) = g(v)$. Then on each fiber, $\tilde{g}_x$ is an isomorphism and this map is continuous, by 2.1, $\tilde{g}$ is a vector bundle isomorphism. $\square$

Now if $f : X \to Y$, one can define $\mathrm{Vect}(f)([F]) = [f^*F]$ and this is well-defined by the previous proposition. Since $\mathrm{Vect}(f \circ g) = \mathrm{Vect}(g) \circ \mathrm{Vect}(f)$ and $\mathrm{Vect}(\mathbb{1}_X) = \mathbb{1}_{\mathrm{Vect}(X)}$, $\mathrm{Vect} : \mathbf{Top} \to \mathbf{CRig}$ is a contravariant functor. We will denote $\mathrm{Vect}(f)$ as f^* when no confusion arises.

Proposition. *If* $f : X \to Y$ *is a homotopy equivalence,* $f^* : \mathrm{Vect}(Y) \to \mathrm{Vect}(X)$ *is a semiring isomorphism.*

Proof. Let $g : Y \to X$ be the homotopy inverse of f, $g \circ f \simeq \mathbb{1}_X$ and $f \circ g \simeq \mathbb{1}_Y$ implies $f^* \circ g^* = \mathbb{1}_{\mathrm{Vect}(X)}$ and $g^* \circ f^* = \mathbb{1}_{\mathrm{Vect}(Y)}$ by 2.6, so f^* is an isomorphism. $\square$

3.2 The Ringification of Semirings

Definition 3.2. Let $S \in \mathbf{CRig}$, the ringification of a semiring S is a commutative ring $R(S)$ with a semiring homomorphism $\imath_S : S \to R(S)$ satisfying the universal property that for any commutative ring R and semiring homomorphism $\varphi : S \to R$, there's a unique ring homomorphism $\tilde{\varphi} : R(S) \to R$

so that $\tilde{\varphi} \circ \imath_S = \varphi$, i.e., the following diagram commutes:

$$\begin{tikzcd} S \arrow[r, "\imath_S"] \arrow[d, "\varphi"] & R(S) \arrow[dl, dashed, "\tilde{\varphi}"] \\ R & \end{tikzcd}$$

Proposition. *There exists such a construction.*

Proof. $(S, +)$ is a monoid, so one can extend $(S, +)$ to an abelian group by the following construction. Define $(S \times S, +)$ by $(a, b) + (a', b') = (a + a', b + b')$, then the binary operation is commutative and $(0, 0)$ is the identity, so $(S \times S, +)$ is a commutative monoid.

Define $(a, b) \sim (a', b')$ on $S \times S$ if there exists some $k \in S$ so that $a + b' + k = a' + b + k$ and denote $R(S)$ to be equivalence classes of this relation and denote the equivalence class of (a, b) as $a - b$. Let $(a, b), (a', b')$ be two representatives of $a - b$ and $(c, d), (c', d')$ be those of $c - d$. Then by definition, there exists $k, l \in S$ so that $a + b' + k = a' + b + k$ and $c + d' + l = c' + d + l$. Notice that $(a, b) + (c, d) = (a + c, b + d)$ and $(a', b') + (c', d') = (a' + c', b' + d')$ and $a + c + a' + c' + k + l = b + d + b' + d' + k + l$, so $(a + c, b + d) \sim (a' + c', b' + d')$, i.e., addition by $(a - b) + (c - d) = (a + c) - (b + d)$ is well-defined. Addition is also well-defined and $0 := 0 - 0$ is the identity. For $a - b \in R(S)$, $b - a \in R(S)$ and $(a - b) + (b - a) = (a + b) - (a + b) = 0$, so each element in $R(S)$ has an inverse, hence $(R(S), +)$ is an abelian group.

Define the product by $(a - b)(c - d) = (ac + bd) - (ad + bc)$, then one can verify that this product is well-defined, distributive, and commutative with identity $1 := 1 - 0$. Hence $(R(S), +, \times)$ is a ring. Notice that we have the semiring homomorphism $\iota_S : S \to R(S), s \mapsto (s, 0)$. One can also verify that $(R(S), \iota_S)$ satisfies the universal property. $\square$

Proposition. *Let $f : S \to T$ be a semiring homomorphism. Then $R(f) : R(S) \to R(T), a - b \mapsto f(a) - f(b)$ is a well-defined ring homomorphism satisfying the following commutative diagram*

$$\begin{tikzcd} S \arrow[r, "\imath_S"] \arrow[d, "f"] & R(S) \arrow[d, "R(f)"] \\ T \arrow[r, "\imath_T"] & R(T) \end{tikzcd}$$

Corollary 3.1. $R : \mathbf{CRig} \to \mathbf{CRing}$ *is a covariant functor.*

3.3 The Functor $K : \mathbf{Top} \to \mathbf{CRing}$ and $\widetilde{K} : \mathbf{Top}_* \to \mathbf{CRng}$

Define $K : \mathbf{Top} \to \mathbf{CRing}$ as the contravariant functor by the composition $R \circ \mathrm{Vect}$. We will denote $K(f)$ as f^* when no confusion arises.

Theorem 3.1. *If $f : X \to Y$ is a homotopy equivalence, then $K(f) : K(Y) \to K(X)$ is a ring isomorphism.*

Proof. By 3.1, $f^* : \mathrm{Vect}(Y) \to \mathrm{Vect}(X)$ is an isomorphism, so $K(f) = R(f^*) : K(Y) \to K(X)$ is an isomorphism by functoriality. $\square$

Example 3.2.1. When X has a homotopy type of a point, $\mathrm{Vect}(X) = \{\epsilon^0, \epsilon^1, \epsilon^2, ...\} \cong (\mathbb{N}, +, \times)$, so $K(X) = (\mathbb{Z}, +, \times)$.

Let $\mathbf{Top}_*$ denote the category of pointed topological spaces. Fix $(X, x_0) \in \mathbf{Top}_*$.

Definition 3.3. Let $\iota_{x_0} : \{x_0\} \to X$ denote the inclusion map of x_0. Then $\iota_{x_0}^* : K(X) \to K(\{x_0\})$ is a ring homomorphism. Define $\widetilde{K}(X, x_0) = \mathrm{Ker}(\iota_{x_0}^*)$.

Proposition. *Let $X, Y \in \mathbf{Top}$, there exists a unique ring homomorphism $\widetilde{K}(f) : \widetilde{K}(Y) \to \widetilde{K}(X)$ so that the following diagram with rows being exact sequence commutes:*

$$\begin{tikzcd}
0 \arrow[r] & \widetilde{K}(X) \arrow[r] & K(X) \arrow[r, "K(\iota_{x_0})"] & K(x_0) \arrow[r] & 0 \\
0 \arrow[r] & \widetilde{K}(Y) \arrow[r] \arrow[u, "\exists! \widetilde{K}(f)"] & K(Y) \arrow[r, "K(\iota_{y_0})"] \arrow[u, "K(f)"] & K(y_0) \arrow[r] \arrow[u, "K(f|_{x_0})"] & 0
\end{tikzcd}$$

This makes $\widetilde{K}$ a functor.

Proof. Because $y_0 = f \circ \iota_{x_0} = \iota_{y_0} \circ f$, by functoriality of K, the right sqaure commutes. Define for $[E] - [F] \in \widetilde{K}(Y)$, $\widetilde{K}(f)([E] - [F]) = K(f)([E] - [F])$, then $K(\iota_{x_0}) \circ \widetilde{K}(f)([E] - [F]) = K(\iota_{y_0}) \circ K(\iota_{x_0})([E] - [F])$, so $\widetilde{K}(f)$ is a map to $\widetilde{K}(X)$. The commutativity of the left square translates to $\widetilde{K}(f) = K(f)$, so the uniqueness is ensured. Functoriality follows by adding another sequence to the top and using the functoriality of K. $\square$

Denote $\Sigma : \mathbf{Top}_* \to \mathbf{Top}_*$ as the covariant functor of reduced suspension. One can define $\Sigma^n : \mathbf{Top}_* \to \mathbf{Top}_*$ as the n-fold composition of Σ.

Definition 3.4. Let $\widetilde{K}^{-n} := \widetilde{K} \circ \Sigma^n : \mathbf{Top}_* \to \mathbf{Top}_*$, then $\widetilde{K}^{-n}$ is a contravariant functor.

4 Brouwer Fixed Point Theorem

Let X, Y be topological spaces. Consider the *external product*

$$\mu : K(X) \otimes K(Y) \to K(X \times Y)$$
$$a \otimes b \mapsto a * b := p_1^*(a) p_2^*(b)$$

where $p_i, i = 1, 2$ are the projection maps onto the ith coordinates.

Proposition. *[2, p.41] The external product is a ring homomorphism.*

One has then a ring homomorphism $\mu : K(X) \otimes K(S^2) \to K(X \times S^2)$.

The relation $(H \otimes H) \oplus 1 \cong H \oplus H$ holds in $K(S^2)$, so the ring homomorphism $\mathbb{Z}[H] \to K(S^2)$ descends to $\mathbb{Z}[H]/(H-1)^2 \to K(S^2)$.

Combining the above two maps we have for any space X the ring homomorphism μ

$$K(X) \otimes \mathbb{Z}[H]/(H-1)^2 \to K(X) \otimes K(S^2) \to K(X \times S^2)$$

Theorem 4.1 (The Fundamental Product Theorem)**.** *[2, p.41, theorem 2.2] μ is an isomorphism.*

Take $X = \{pt\}$, then $K(S^2) \cong \mathbb{Z}[H]/(H-1)^2$ and $\widetilde{K}(X) = \{a(H-1) : a \in \mathbb{Z}\}$, which is isomorphic to $\mathbb{Z}$.

The external product descends to the reduced K-theory $\mu : \widetilde{K}(X) \otimes \widetilde{K}(Y) \to \widetilde{K}(X \wedge Y)$. Let $X = S^2$, then $\mu : \widetilde{K}(S^2) \otimes \widetilde{K}(Y) \to \widetilde{K}(\Sigma^2 Y)$

Theorem 4.2 (Bott's Periodicity Theorem)**.** *Define the following rng isomorphism*

$$(H-1) \otimes \cdot : \widetilde{K}(Y) \to \widetilde{K}(S^2) \otimes \widetilde{K}(Y)$$
$$a \mapsto (H-1) \otimes a$$

Then $\mu \circ ((H-1) \otimes \cdot) : \widetilde{K}(Y) \to \widetilde{K}(\Sigma^2 Y)$ is an isomorphism of rng isomorphisms..

Corollary 4.1. *$\tilde{K}(S^{2n}) \cong \mathbb{Z}$, in particular the $\mathbb{Z}$-module in $K(S^{2n})$ on the basis $(H-1)*...*(H-1)$(n-fold). $\tilde{K}(S^{2n+1}) = 0$. In particular, $\widetilde{K}^*(S^n)$ is non-trivial for some $*$.*

Corollary 4.2 (Brouwer Fixed Point Theorem)**.** *A continuous map $f : D^n \to D^n$ has at least one fixed point.*

Proof. Define $r(\alpha, x) = (1-\alpha)f(x) + \alpha x$, where $\alpha > 0$. Then r is a continuous function over x and α. Fix $x_0 \in D^n$, then as α goes from 0 to infinity, $r(\alpha, x_0)$ is a ray from $f(x_0)$ to x_0. Since $f(x_0) \neq x_0$, this ray must intersect S^{n-1} once, therefore there's a unique α_0 so that $|r(\alpha_0, x_0)| = 1$. Define in this way for each $x \in D^n$ an $\alpha(x)$.

Because r is continuous, $|r|$ is continuous, so $|r|^{-1}(1)$ is a closed set. Let x_n be a sequence of point in D^n so that $x_n \to x$, then since $(\alpha(x_n), x_n) \subset |r|^{-1}(1)$, $(\alpha(x_n), x_n) \to (\alpha, x) \subset |r|^{-1}(1)$, by uniqueness, $\alpha(x) = \alpha$, so $\alpha(x_n) \to \alpha(x)$, proving $\alpha(x)$ is continuous over $x \in D^n$.

If $x \in S^{n-1}$, $\alpha(x) = 1$ because the ray starting from $f(x)$ to x intersect S^{n-1} at x when $\alpha = 1$, so $r \circ \iota = \mathbb{1}_{S^{n-1}}$, where ι is the inclusion of S^{n-1} into D^n.

Hence by functoriality of $\widetilde{K}^{-m}$, $\widetilde{K}^{-m}(\iota) \circ \widetilde{K}^{-m}(r) = \widetilde{K}^{-m}(\mathbb{1}_{S^{n-1}}) = \mathbb{1}_{\widetilde{K}^{-m}(S^{n-1})}$, hence $\widetilde{K}^{-m}(\iota)$ is a monomorphism. Since D^n deformation retracts to a point, $\widetilde{K}^{-m}(D^n) \cong 0$, so $\widetilde{K}(\iota) : \widetilde{K}^{-m}(S^{n-1}) \to \widetilde{K}^{-m}(D^n)$ is injective only if $\widetilde{K}^{-m}(S^{n-1}) \cong 0$. But $\widetilde{K}^{-m}(S^{n-1}) = \widetilde{K}(\Sigma^m S^{n-1}) \cong \widetilde{K}(S^{m+n-1})$ is nontrivial for some m by 4.1. $\square$

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